

HPAS Design Decision Justification

Justification of Decisions — Hospital Patient Referral, Treatment and Administration System

Document Version	v1.0	Related Documents	HPAS Acceptance Test Plan (Acceptance Test Plan) HPAS-JOURNEY. BPMN (process map) + 8 sub-processes
Decision Subject	9 BPMN models (17 process definitions), 39 Camunda Forms, and 31 Job Types	Supporting Illustration	
Scope of Argumentation	Process structure / Participant boundaries / Task allocation / Gateways / External interactions / Exception handling / Assumptions / Alternatives / Trade-offs	Evidentiary Basis	All decisions have been validated in a local Camunda 8 cluster (including the actual outcomes of rejected proposals).

0. How to Read

Each decision should be articulated in four paragraphs, avoiding the presentation of only the conclusion:**Decision** → **Rationale** → **Alternative Options Considered** → **Trade-offs and Costs**. Section 10 consolidates the alternative options and trade-offs into two tables, while Section 11 provides a traceability mapping of "Decision → Model Elements → Acceptance Test Cases."

1. Process Structure

For DD-01, use "one main process + 8 reusable sub-processes" instead of "one large flowchart" or "8 parallel choices."

Decision Overall Layout Process_HPAS_Journey Using 8 callActivity(zeebe: called Element processId=...) connect the 8 options end-to-end:① P1 Referral Verification → ② P2 Clinical Decision → ③ P3 Scheduling → ④ P4 Notification → ⑤ P10 Letter → ⑥ P7 Funding → ⑦ P12 Payment → ⑧ P8 Consultation Routing.

Reason The case describes a single path **for the same patient, traversed along a timeline**, rather than eight independent scenarios. Using Call Activity to connect these steps can simultaneously achieve three objectives:① the overall diagram answers the question "What is the overall workflow?";② each sub-process can be independently deployed, versioned, and tested;③ sub-processes can be reused elsewhere (e.g., the "Follow-up Appointment" process could also reuse P3 scheduling).

Other options are shown in Section 10: A-1 (one large flowchart), A-2 (eight parallel choice branches), and A-3 (embedded sub-processes).

judge and weigh The trade-off is increased instance depth: at runtime, one must expand **Called Instances** within the Operate section to view the internals of sub-processes. Mitigation: use a swimlane overview in the overall diagram combined with documentation for each Call activity to specify the key rules and external parties for that phase; limit the number of sub-processes to 8 to keep the cognitive load under control.

evidence Actual end-to-end measurement: total instances 2251799813750025= COMPLETED, all 8 sub-instances executed in correct order and are COMPLETED, incident = 0.

The sub-processes of DD-02 are connected via "variable pass-through + one explicit decision gate," rather than relying on manual memory.

Decision For all call activities, set propagateAllChildVariables= "true",and the decision result of the sub-process (isComplete,reviewForm.decision,scheduleStatus,paymentStatus...)Automatically pass back to the parent process; in the overall flowchart, after ② place a unique decision gate "Does the consultant physician accept?" (XOR, condition= reviewForm!= null and reviewForm.decision = "accept").

Reason The hard constraint of this case is that "*A new patient appointment may only be arranged after an authorised Consultant has accepted the referral*".This constraint spans the boundaries of sub-processes and must be enforced by a **reviewable and testable** gateway, rather than being distributed across various sub-processes via informal agreements.

Alternative Approaches: Leave the determination of "Accept/Reject" to the P3 sub-process for independent verification (Rejection criteria: constraints are hidden; cannot be visually identified on the overall layout); or rely on workflow variables for manual judgment (Rejection criteria: no audit trail).

judge and weigh The parent process requires knowledge of the variable names in the child process (reviewForm.decision), which introduces a certain degree of coupling. Mitigation: This field is a business primary key-level convention; the documentation already integrated into the model, along with the test plan TC-05, addresses this issue.

2. Participant Boundaries

For intra-organizational use, use "lane"; for inter-organizational use, use "Pool + messageFlow".

- Decision** Internal hospital teams are represented as **five swimlanes within the same Pool**(medical secretary / consultant physician / outpatient appointment team / call handling / finance team); external parties are represented as **independent external Pools**, with only messageFlow permitted between the external parties and the hospital; sequence flows are not allowed to cross Pool boundaries.
- Reason** This illustrates the semantic roles in BPMN:*Pool* represents independent participants (different legal entities / different systems), whose interactions consist of **asynchronous, deferrable, and failable** messages;*Lane* represents the division of responsibilities within the same organization, whose interactions consist of sequential flows within a process. In this case, external parties (GP, scheduling service, communication service, funding agency, PSP) are all located outside the hospital; representing them in separate pools could obscure the real risk that "messages may be lost and acknowledgments may be delayed."
- Other options** are described in Section 10: A-4 (External Parties using shared lanes), A-5 (Black-box Pool), and A-6 (External Parties using separate pools).
- judge and weigh** The dual-pool approach increases the height of each sub-flowchart and the number of nodes. The mitigation strategy is to include the five swimlanes of the external parties in a single pool within the overall flowchart, using dashed message flows to summarize "with whom interactions occur in this phase," while keeping the detailed information within the sub-flowcharts.
- evidence** In actual testing, the cross-span Pool message did indeed trigger external party instances:P2_PartnerPool, P3_SchedulingPool (COMPLETED)、 P4_PatientPool 、 P12_PSPPool (COMPLETED)。

Create an "experimental" process for the external Pool DD-04 with isExecutable= "false", while completing the process start/end points and interconnections.

- Decision** The external pool references a non-executable process (5 swimlanes, 6 nodes),**which likewise features a start event, an end event, and a fully connected sequential flow**; the node types utilize userTask (human actors: referring party / patient / funding agency), natively supported by Camunda 8, and serviceTask+zeebe:taskDefinition(system actors: scheduling / communication / PSP).
- Reason** ① External parties shall not create a process in the Tasklist that can be arbitrarily triggered → use a non-executable type;② however, modeling specifications require each process to have start and end events, and Camunda 8 does not support the generic bpmn:task → the required structure must be completed; otherwise, the Problems panel in Camunda Modeler will throw an error (see the actual records in Section 8, D-02/D-03).
- Other Options** Black-box Pool (no processRef) — Rejected: Cannot be attached to a泳道; cannot serve as a message stream endpoint to represent internal operations of an external party. Allow external Pool execution — Rejected: Would pollute the list of executable processes, and the external party is not driven by this instance.
- judge and weigh** A "non-executable" process has been added to the model; its purpose should be specified in the documentation. Mitigation: In the documentation of the process, specify that it is "for modeling purposes only and does not participate in execution."
- evidence** Deployment in actual environment:Process_External_Parties was ignored by Zeebe, and only executable processes were deployed; Modeler validation status changed from "Error" to "No Issues".

3. Task Allocation

DD-05: Classify User Tasks and Job Workers based on "Who Makes the Decision / Who Performs the Computation"

Decision	User Task Involves performing human-dependent actions and generating corresponding records: registering referrals, conducting clinical review and decision-making, determining funding sources, granting/denying approvals, performing administrative inspections, following up on outcomes, handling payment anomalies, and responding to inquiries. Service Task (Job Worker) handles deterministic computations and system calls: material integrity verification, grant validity validation, cost calculation, external interface calls, message delivery, and audit logging.
Reason	The case repeatedly emphasizes the separation of duties and auditability: clinical decisions must be made by authorized clinical personnel; administrative personnel may only perform administrative checks; and financial decisions must be made by financial personnel. These actions should be defined as User Tasks and restricted to candidateGroups. Defining candidate groups effectively embeds the "responsibility boundaries" into an executable model, rather than merely documenting them in the requirement specification. Alternative approaches are described in Section 10: A-7 (creating User Tasks entirely) and A-8 (delegating all decision logic to Workers).
judge and weigh	User tasks require manual intervention, which slows down end-to-end execution (the main path of the 8 sub-processes in this project involves a total of 16 manual tasks). Mitigation: Fully automate all pure computational tasks and API calls, retaining only those nodes that require human intervention; during demonstrations, use scripts or Tasklists to advance tasks in batches.

The DD-06 candidate groups are named based on "teams in the case" rather than "system roles".

Decision	candidateGroups Original team names for use cases: medical-secretaries, consultants, appointment-team, call-handling, finance-team, admin-managers, clinical team, gp-practices, funder-reviewers, communication-service, referring practices.
Reason	Ensure that the model's "track/candidate group" corresponds one-to-one with the "organizational unit" of the case, allowing reviewers to verify the correspondence against the original case text; It also facilitates future integration with real organizational structures (LDAP/IdP group mapping).
judge and weigh	If an organization changes its name, its model must also be updated; using a role name (e.g., referral-intake-role) can decouple the system, but at the expense of readability. In the current phase, readability is prioritized.

4. Gateways

For DD-07, business logic decision-making uniformly employs an Exclusive OR (XOR) gate combined with a default branch.

Decision	All business rules utilize exclusiveGateway; each outbound sequential flow either includes a conditionExpression or is declared as the default of the gateway. The default branch always points to the "Convergence/Recording" path (e.g., "Record exception (duplicate/incomplete)", "Notification send failure", "Referral not accepted"), rather than being silently discarded or treated as a technical error.
Reason	① These judgments are mutually exclusive in business terms and have a well-defined priority order; ② explicit defaults ensure that "no conditions were matched" also has a definitive outcome (fail-safe); ③ Camunda 8 requires every outgoing flow of a gateway to have either a condition or a default; otherwise, model validation fails.
Other solutions	include: Inclusive gateway: No "satisfying multiple conditions simultaneously" criteria are present in the given case; Parallel gateway: These criteria cannot be executed in parallel; Decision table (DMN): Suitable when the number of rules is small and each rule must correspond one-to-one with a process node; DMN introduces additional components.
judge and weigh	An increase in the number of gateways can increase diagram complexity (e.g., P12 has three state branches). Mitigation: name branches directly using business semantics ("Success", "Failure/Rejection", "Repeat/Incomplete") to improve readability for business stakeholders.

The DD-08 gateway must perform a security evaluation of "dirty data".

Decision	Conditional expressions should not directly call user-input functions such as <code>date()/number()</code>, which may throw exceptions. For example, replace the "Has the appointment been made within two weeks?" in P4 with the following: <pre>= appointmentDate != null and appointmentDate != "" and appointmentDate <= string(today() + duration("P14D"))</pre> Additionally, the form applies a regular expression validation <code>^\d{4}-\d{2}-\d{2}\$</code> to the date field.
Reason	Actual defect D-01: When the appointment date is set to 123, the parsing of <code>date("123")</code> fails and returns null; consequently, XOR cannot retrieve a boolean value → triggers a production incident (EXTRACT_VALUE_ERROR). Defenses should be implemented in two layers: entry validation (forms) to ensure data integrity, and expression fault tolerance to prevent process interruption.
judge and weigh	String comparisons rely on ISO formatting (lexicographical order = chronological order); dates without leading zeros may lose their validity, so this must be handled by a form regular expression. If the input source becomes uncontrolled in the future (e.g., directly transmitted from an external system), normalization should be performed within the Worker before further validation.

5. External Interactions

For DD-09, cross-Pool interactions must always use "message + association key"; synchronous calls are not permitted.

- Decision** Steps requiring external participation are uniformly replaced as follows:**Throw message**(intermediateThrowEvent or endEvent+zeebe:taskDefinition, delivered by the Worker) → **Capture acknowledgment**(intermediateCatchEvent+ bpmn: message with zeebe: subscription correlationKey). The association key is set to the business primary key:referralId(referral verification / clinical decision / scheduling),appointmentId(funding approval),caseId(payment).
- Reason** The case study explicitly lists real-world scenarios of "communication failure / no response received" (*payment is successfully taken... but confirmation is not returned due to a communication failure*). Synchronous calls cannot represent an intermediate state of "sent, no response received"; whereas a combination of messaging and a waiting node naturally captures this state, aligning with the actual integration approach with external parties (asynchronous interfaces / manual response handling).
- Alternative solutions** are described in Section 10, A-9 ("Synchronous invocation of external systems within Service Tasks") — this represents our first implementation attempt; while it has been successfully validated in practice, it does not support handling cases such as lost acknowledgments, external system unavailability, or manual acknowledgments.
- judge and weigh** For each message, design a correlation key, a TTL (currently 10 minutes), and a retry delivery strategy; add the following test step: (wait + trigger acknowledgment). Mitigation: reuse the business primary key as the correlation key to avoid introducing an additional correlation field.

evidence Actual measurements: For P3, both scheduling requests and results; for P7, funding approvals; and for P12, payment requests and receipts—all were handled via external queue instances, from initiation to completion.

For the message name and the associated key, refer to Section 3.2 of the acceptance test plan.

For the DD-10, external components are summarized on the overall layout using "dashed data flows + swimlanes," without duplicating internal details.

- Decision** In the external Pool of the master diagram, only six "touchpoint" nodes are placed (providing referral materials / allocating resource slots / sending notifications and correspondence / approving funding / processing payments / handling patient calls), and eight messageFlow connections are established to the corresponding Call activities at each stage.
- Reason** The purpose of the overall diagram is to answer "which external parties the entire system interacts with, and at which stages," rather than reiterating the message details of each external interface. Such details are already provided in the dual-pool diagrams of each sub-process.
- judge and weigh** The message flow in the overall layout is a "summary view" rather than individual messages; strictly speaking, it does not directly correspond to any specific message in runtime.*If the review requirements demand strict one-to-one correspondence*, it may be modified by omitting the message flow and retaining only the external Pool (i.e., adopting the simplified approach described in A-5).

6. Exception Handling

DD-11: Business exceptions are represented using BPMN branches; technical exceptions are represented using BPMN Error / Retry.

Decision	In this case, all anomalies are handled via explicit branches and manual nodes: missing item loop (request replacement → wait for return → re-verify), no available slot in scheduling (remain in Pending state + trigger Timer-based retry), notification failure (after ≤ 3 retries, transition to "notification delivery failed"), payment failure / duplicate / incomplete payment (each of these three branches is handled manually), and overdue mail (escalate to an administrative manager / management level). Technical failures of jobs are still handled by zeebe:taskDefinition retries and the retry mechanism of Operate.
Reason	These "abnormalities" are, in business terms, actually normal pathways that require traceability and manual approval (BR-10 / BR-17 / BR-19 / BR-27) If a BPMN Error Is Raised, It Will Bypass the Audit Trail Form and Be Presented as an "incident," Effectively Misclassifying a Business Event as a Technical Failure.
judge and weigh	The number of branches has increased significantly (P4 features notification-based retry + two-week evaluation; P12 includes three-state branching). Mitigation: all default branches converge to the "Record / End" node, preventing the workflow from becoming orphaned.

DD-12 Acceptance Criterion: Abnormal paths must not be exposed as incidents.

Decision	Incorporate the requirement "During execution, incident = 0" into the acceptance criteria (AC-5), and mandate that each exception branch have a corresponding test case.
Reason	An "incident" indicates that the workflow has stalled and requires manual intervention in the Operate stage; such incidents are classified as operational incidents. In contrast, the case scenario requires the system to "detect delays and failures" and continue progressing. Treating the resolution of incidents as a measurable acceptance criterion can, in turn, serve as a feedback mechanism to enhance modeling quality.
evidence	D-01: After repair, retest using the same dirty data; incident = 0 (acceptance test plan TC-20).

7. Assumptions hypothesis

In cases where the requirements document does not provide sufficient information, the values specified in the following table shall be applied, and the resulting impacts shall be documented; these assumptions shall also be recorded in Section 3.2 of the acceptance test plan to facilitate unified revisions following business confirmation.

Number	Hypothesis	Basis / Impact (if the assumption does not hold)
AS-01	"Must be contacted by phone within two weeks" is determined based on whether the appointment date $\leq$ today + 14 days .	For the case, specify only <i>within the following two weeks</i> . If this is changed to "working calendar days," the gateway conditions and the test data for TC-08 must be adjusted accordingly.
AS-02	Letter processing timeline: Drafting → Approval → Administrative Review → Distribution Total target: 7 days ; delays beyond this period will trigger escalation.	The case study defines two escalation levels: 7 days and 1 month or 3 months; this model implements the first level (administrative manager), while higher levels are documented in separate specifications.
AS-03	If the notification fails, it will be retried up to 3 times ; otherwise, it will be terminated and a record will be retained.	No limit is set on the number of attempts in this case. Use three attempts to balance delivery probability with avoiding unlimited 重试; this can be adjusted simply by modifying the gateway conditions.
AS-04	Cross-Pool message TTL = 10 minutes ; no timeout escalation for waiting nodes	For demonstration and local validation purposes; in production, reconfiguration must comply with OQ-01 (cross-Pool processing time limits) and the business agreement.
AS-05	The urgency determination uses the " urgency keywords " as a temporary criterion, which is propagated along with the message and highlighted.	The case clarifies that this rule <i>has not yet reached consensus</i> (further confirmation with stakeholders is required). A provisional approach is adopted to ensure the process remains operational; once the rule is finalized, it will be replaced with a decision table / rule service.
AS-06	Funding approval shall only proceed upon receipt of external feedback ; otherwise, the approval will be automatically granted after a timeout period.	This case highlights that funding serves as a prepayment gate; if timeout-based approval is permitted, additional boundary timing events and authorization records must be implemented.
AS-07	External parties participate in the model via "person + form" or "system interface"; external pools do not participate in actual execution.	The scope of this project covers the process orchestration of the on-premises systems; external processes of third parties are not included in the delivery scope.
AS-08	Process ID / Variable Name (referralId,appointmentId, caseId)serves as the business primary key	Once the association key is changed, all cross-Pool messages and test data must be synchronized.

8. Issues Identified During Decision Implementation (empirical Evidence Supporting the Rationale)

Number	Phenomenon	Impact on design
D-01	Gateway expression date(appointmentDate) Encountering invalid input 123 Throwing an incident	Ensure DD-08: conditional expressions must be null-safe + form pre-validation; and fulfill acceptance criterion AC-5 (incident = 0).
D-02	Non-executable external pool in Camunda Modeler reports <i>Process is missing start event</i>	Enforce DD-04: Non-executable processes must also have a complete structure (start and end events + connectivity) and must not use the generic bpmn:task model, which is not supported by Camunda 8.
D-03	Option 1: Remaining isolated nodes within the model Activity_1 kuvs54	Facilitate the inclusion of the "Structure Checker" in the release: ID uniqueness, referential integrity, full DI compliance, start and end events, isolated nodes, and element type compliance

		(Approved on 9/9).
D-04	In earlier versions, the use of "Zeebe embedded forms" forced user tasks to adopt a job-worker model, which prevented Tasklist from launching workflows.	Facilitate "User Task Uniform External Link form" (zeebe: formDefinition formId) + "No Type" launch event + "Launch Form" — this unified approach ensures that business users can initiate processes via self-service within the Tasklist.

9. Summary of Alternative Options and Trade-offs

9.1 Other Alternatives Considered

Number	Alternative	Reasons for withdrawal / When a re-election should be held
A-1	Draw the eight options onto a single large plane.	The map sheet is too large, cannot be reused, and cannot be provided as an independent version; the responsibility boundaries are not clearly defined.If the deliverable requires only a "visual schematic for presentation" rather than an executable model, this option may be considered instead.
A-2	The master diagram is split into eight parallel branches via a gateway (using the "Which process to select" routing).	Already implemented and rejected: it reduces the patient journey to unrelated options, failing to capture the sequential relationships between stages and data handoffs (e.g., "appointment can only be scheduled after acceptance").However, if the requirement is for a "scenario demonstration entry checklist," this approach would be more intuitive.
A-3	Replace the Call activity with an embedded subprocess	It is non-reusable, cannot be deployed independently, lacks versioning support, and the parent diagram cannot collapse its details.If the child process serves only a single parent process and is never reused, this design can be revised.
A-4	Draw the external party as a swim lane within the hospital pool.	This obscures the asynchronous and failure semantics across organizations, reducing "messages" to a sequential stream.If external parties are actually driven synchronously by the same system within this organization, this approach may be replaced.
A-5	The external pool uses a black box (without processRef) and does not depict message flows.	The lanes cannot support external interactions, and message exchange remains invisible.If the review only requires "visibility of participant boundaries," this is the simplest solution.
A-6	Each external party has its own independent pool.	The overall diagram will consist of 6+ pools, significantly reducing readability.If you require per-party SLA / per-party audit views, you may make an alternative selection.
A-7	All nodes have been configured to perform user tasks (manual execution).	Loses automation value, and compute nodes cannot guarantee consistency.Suitable for the prototype review phase, facilitating business stakeholders in clarifying responsibilities.
A-8	Move all Business Rule (BR) evaluations down to Job Worker code.	Rules are not visible in the diagram, cannot be reviewed by the business stakeholders, and are difficult to test in isolation.When the number of rules is large and they are frequently changed, this approach should be replaced with a DMN decision table combined with Business Rule Tasks.
A-9	Synchronous invocation of external systems using Service Tasks (first implementation version)	Already implemented and replaced: This option cannot be used for scenarios such as "messages have been sent but no acknowledgment has been received," "external system temporarily unavailable," or "messages manually returned."If the external interface supports strong consistent synchronous communication without latency, this option may be considered instead.
A-10	Cross-Pool messages use embedded forms in Camunda 8 to carry pending tasks.	Actual pitfalls identified and rejected: embedding forms converts user tasks into job-worker type, which prevents Tasklist from initiating workflows (D-04).
A-11	Use the BPMN Error event to represent business exceptions such as missing items or payment failures.	This approach may represent business events as technical failures, thereby bypassing audit trails and manual decisions.Genuine technical failures (e.g., interface timeouts, system unavailability) should still be represented using BPMN Error.

9.2 Trade-Offs Between Trade-off Options

Trade-off dimensions	Trade-offs of this project	Costs and Mitigation
Readability ↔ Fidelity	Bias toward fidelity: retain dual pools and all anomaly branches	The sub-flowcharts are more complex. To simplify: the master diagram uses Call Activity folding combined with lane overview; each individual diagram focuses on a single stage.
Reusability ↔ End-to-end visibility	Reusability-oriented: 8 independent sub-processes + overall diagram orchestration	During operation, deployed Called Instances shall be activated. Mitigation: the overall diagram swimlane provides a global view; the test plans TC-01/TC-02 specify verification of the call chains.
Automation ↔ Auditable	Computing and interface automation, along with decision-making and traceability, are essential for human operations.	High number of manual tasks and lengthy demonstration time (16 tasks). Mitigation: Workers automatically complete the remaining steps; demonstrations can be scripted and executed efficiently.
Strict validation ↔ Operational fault tolerance	Two-layer defense: form validation (entry point) + expression null-safety (runtime)	The expression employs string comparison and relies on ISO formatting conventions. Mitigation: Incorporate these conventions into the test data baseline and DD-08.
Asynchronous messaging ↔ Test complexity	Asynchronous selection: capable of capturing authentic failure patterns	Each message must be associated with a association key, a TTL, and a waiting node; the test should simulate acknowledgments. Mitigation: reuse the business primary key for the association key; the acceptance test plan should specify fixed test data.
Model completeness ↔ Simple deployment	The external pool is retained as a complete but non-executable process.	Add an additional "non-execute" process. Mitigation: annotate the purpose using documentation; during deployment, only release executable processes.

10. Decision Traceability (Decision → Model Element → Acceptance Test Case)

Decision	Placement Location (Model Element / File)	Verification test cases
DD-01	hpas-journey. bpmn of Call_01_ Referral Intake...Call_08_Enquiry (zeebe: called element)	TC-01、 TC-21
DD-02	Gateway_Accepted+the propagateAllChildVariables of each Call activity	TC-01、 TC-05
DD-03	For each sub-process:participant+messageFlow (e.g.,MessageFlow_SupplementReply)	TC-03、 TC-06、 TC-08、 TC-18
DD-04	Process_External_Parties(isExecutable= "false", 5 lanes + start/end events)	TC-21
DD-05	User Task: Task_RegisterReferral 、 Task_ClinicalReview ...; Service Task: Task_CompletenessCheck 、 Task_ValidateFunding ...	TC-02、 TC-10、 TC-22
DD-06	For each User Task zeebe:assignmentDefinition	TC-22

	candidateGroups	
DD-07	Gateway_Complete, Gateway_NotificationStatus, Gateway_PaymentStatus, and others (all with a default value)	TC-05、TC-09、TC-15、TC-16
DD-08	P4 Gateway_WithinTwoWeeks condition +hpas-start-journey/p4-start appointment form regular expression	TC-20
DD-09	bpmn: message+zeebe: subscription correlationKey; on throw/end events zeebe: taskDefinition	TC-03、TC-06、TC-12、TC-14
DD-10	The six contact point nodes of the external Pool in the master layout are connected to eight messageFlow edges.	TC-01、TC-21
DD-11	Catch_RetryTimer (PT5S) 、 Flow_NotifyFailed 、 Task_RecordException 、 Task_Escalate	TC-07、TC-09、TC-11、TC-16
DD-12	Acceptance Criteria AC-5; Structural Validator_validate.ps1	TC-20、TC-21

Note: This document and the "HPAS Acceptance Test Plan" are mutually referenced — this document explains "why this design was adopted," while the test plan verifies "whether the design meets the expected objectives." Both documents share the same numbering scheme (TC-xx / BR-xx / AC-x / D-xx) and may be cross-referenced.